

A chromosome consists of a DNA molecule packed together with proteins

Now let's examine how DNA is packaged into chromosomes, the structures that carry genetic information. The main component of the genome in most bacteria is a double-stranded, circular DNA molecule that is associated with specific proteins. A bacterial chromosome differs from a eukaryotic chromosome in that a eukaryotic chromosome consists of a

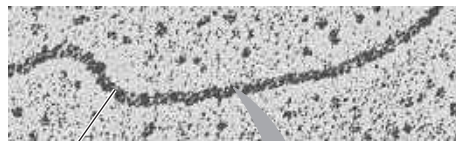
single linear DNA molecule associated with a large number of proteins. In *E. coli*, the chromosomal DNA consists of about 4.6 million nucleotide pairs, representing about 4,400 genes. This is 100 times more DNA than is found in a typical virus, but only about one-thousandth as much DNA as in a human somatic cell. Even so, that is a tremendous amount of DNA to be packaged in such a small container.

Stretched out, the DNA of an *E. coli* cell would measure about a millimeter in length, 1,000 times longer than the region it occupies in the cell. Within a bacterium, however, certain proteins cause the chromosome to coil and “super-coil,” densely packing it so that it fills only part of the cell.

▼ Figure 16.23 Exploring Chromatin Packing in a Eukaryotic Chromosome

DNA, the Double Helix

Shown below is a ribbon model of DNA, with each ribbon representing one of the polynucleotide strands. The TEM shows a molecule of naked (protein-free) DNA; the double helix is 2 nm across.



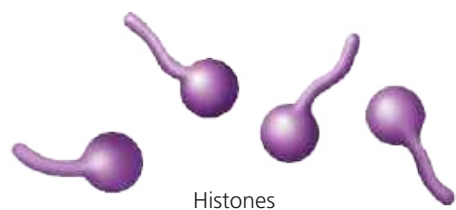
DNA double helix
(2 nm in diameter)



Histones

Proteins called **histones** are responsible for the main level of DNA packing in interphase chromatin. More than a fifth of a histone's amino acids are positively charged (Lys or Arg) and therefore bind tightly to the negatively charged DNA.

Four types of histones are most common in chromatin. The histones are very similar among eukaryotes, probably reflecting their important role in chromatin structure.

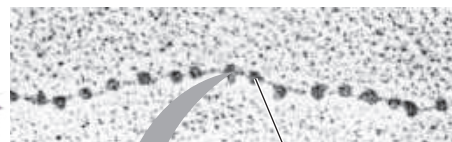


Nucleosomes in a 10-nm Fiber

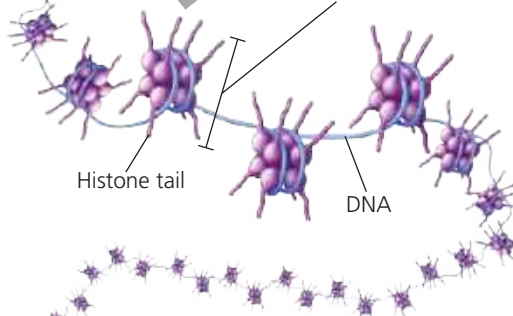
In electron micrographs, unfolded chromatin is roughly 10 nm in diameter (the *10-nm fiber*). Such chromatin resembles beads on a string (see the TEM). Each “bead” is a **nucleosome**, the basic unit of DNA packing; the “string” between beads is called *linker DNA*.

A nucleosome consists of DNA wound twice around a protein core of eight histones, two each of the four main histone types. The amino end of each histone (the *histone tail*) extends outward from the nucleosome and is involved in regulation of gene expression.

In the cell cycle, the histones leave the DNA only briefly during DNA replication. Generally, they do the same during the process of transcription, which requires access to the DNA by transcription proteins.



Nucleosome
(10 nm in diameter)



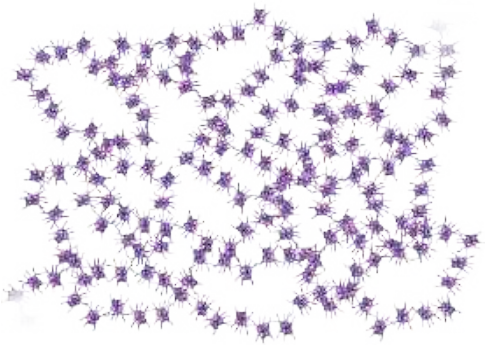
Euchromatin (loosely arranged 10-nm fiber)

Euchromatin/Heterochromatin

A recent technique called ChromEMT allows scientists to visualize chromatin in intact cells. ChromEMT and other new techniques have shown that the 10-nm fiber is the basic constituent of interphase chromatin.

During interphase, different regions of a chromosome may exist as euchromatin (below left) or heterochromatin (below). In euchromatin, the 10-nm fiber is loosely arranged in a more open configuration than heterochromatin; higher degrees of organization, including the 30-nm fiber in one older model, may exist in specific cells or at certain times. (This is an area of very active research.) The DNA in euchromatin is accessible to the proteins that carry out transcription, and its genes can be expressed. In heterochromatin, the 10-nm fiber is more densely arranged and less accessible to these proteins; genes in heterochromatin are generally not expressed.

Euchromatin and heterochromatin are organized into regions by other proteins not shown here; this organization is dynamic but disappears once mitosis begins.



Heterochromatin (densely arranged 10-nm fiber)